Wine Quality Prediction

Regression with physicochemical measurements | Aivancity PGE 2

CLEAN ROWS	TEST RMSE	TEST R-SQUARED
5,320	**0.685**	**0.375**
after duplicate removal	score points	held-out test data

Executive summary

This project estimates the numerical **quality** score of red and white Portuguese Vinho Verde wines from physicochemical measurements. The original expert rating is ordinal, but the assignment is approached as regression. The two UCI files were combined with a wine-type label, then split before model selection. A scikit-learn pipeline keeps imputation, scaling and encoding inside cross-validation. Five regression algorithms are compared against a mean baseline. The final model is selected using training cross-validation and evaluated once on a held-out test split.

Main finding

K-nearest neighbours achieved the lowest cross-validation RMSE (0.715) with 21 neighbours and distance weighting. On the untouched test split, it achieved RMSE 0.685, MAE 0.534 and R-squared 0.375. It outperformed the mean baseline, while leaving substantial variation unexplained.

Dataset: UCI Wine Quality, ID 186 | Red and white Vinho Verde samples | 2 November 2026

Reference: Cortez et al. (2009), UCI Machine Learning Repository, DOI 10.24432/C56S3T. Dataset licence: CC BY 4.0.

1. Dataset and exploration

Item	Observed value
Raw records	6,497
Red / white	1,599 / 4,898
Predictors	11 numeric measurements + wine_type
Missing cells	0
Exact duplicates removed	1,177
Records after cleaning	5,320

The original semicolon-separated UCI files contain 1,599 red and 4,898 white observations. The source type is attached before combining them. Exact duplicates are counted and removed only from the processed copy; the originals remain unchanged. This lowers the chance that identical rows cross the train/test boundary, though two distinct wines could share the same recorded values.

Target and wine-type distributions

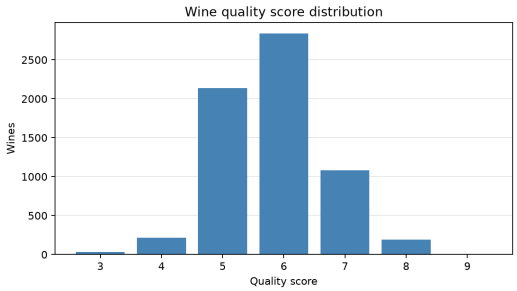


Figure 1. Quality scores in all raw observations.

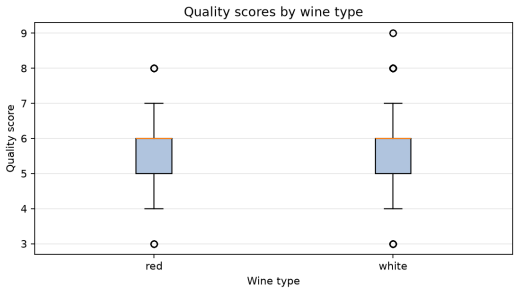


Figure 2. Quality scores by wine type.

Scores are concentrated around 5 and 6, with few observations at the extremes. White wines make up most of the sample, so the pooled pattern is weighted toward them. These plots are descriptive; they do not show that wine type causes quality differences.

2. Method and model selection

Split and preprocessing

The cleaned data were split into 4,256 training rows and 1,064 test rows with a 20% test fraction and seed 42. The test set stayed outside cross-validation and hyperparameter search. A ColumnTransformer uses median imputation and StandardScaler for numerical inputs, plus most-frequent imputation and one-hot encoding for wine_type. Preprocessing is inside each Pipeline, so every fold fits it on that fold's training portion only.

Comparison strategy

The mean DummyRegressor provides a baseline. LinearRegression, Ridge, Lasso, KNeighborsRegressor and LinearSVR use the same shuffled five-fold KFold strategy with seed 42. GridSearchCV tunes meaningful settings for Ridge, Lasso, k-NN and LinearSVR using negative RMSE. The selected model is the candidate with the lowest mean training cross-validation RMSE; test metrics do not choose the model.

Model	CV RMSE	CV MAE	CV R-squared	Selected settings
KNeighborsRegressor	0.715	0.553	0.343	n_neighbors=21, weights=distance
Ridge	0.735	0.567	0.306	alpha=10.0
LinearRegression	0.735	0.567	0.306	no tuned parameters
Lasso	0.735	0.567	0.305	alpha=0.001
LinearSVR	0.737	0.566	0.302	C=10.0
DummyRegressor	0.883	0.697	-0.002	strategy=mean

Metric interpretation

RMSE squares errors before averaging, so large mistakes weigh more. MAE is the average absolute error in quality-score points. R-squared compares squared errors with a mean-prediction reference and may be negative. Since quality is ordinal, regression treats adjacent score gaps as equally spaced; this is an explicit simplification.

3. Final test evaluation

Model on untouched test data	RMSE	MAE	R-squared
DummyRegressor baseline	0.868	0.694	-0.001
KNeighborsRegressor	0.685	0.534	0.375

The selected model reduces test RMSE by 21.0% relative to the mean baseline. Its MAE of 0.534 means the absolute prediction error averages a little over half a score point on this test split. Its R-squared is 0.375: useful predictive signal is present, but much of the variation remains unexplained.

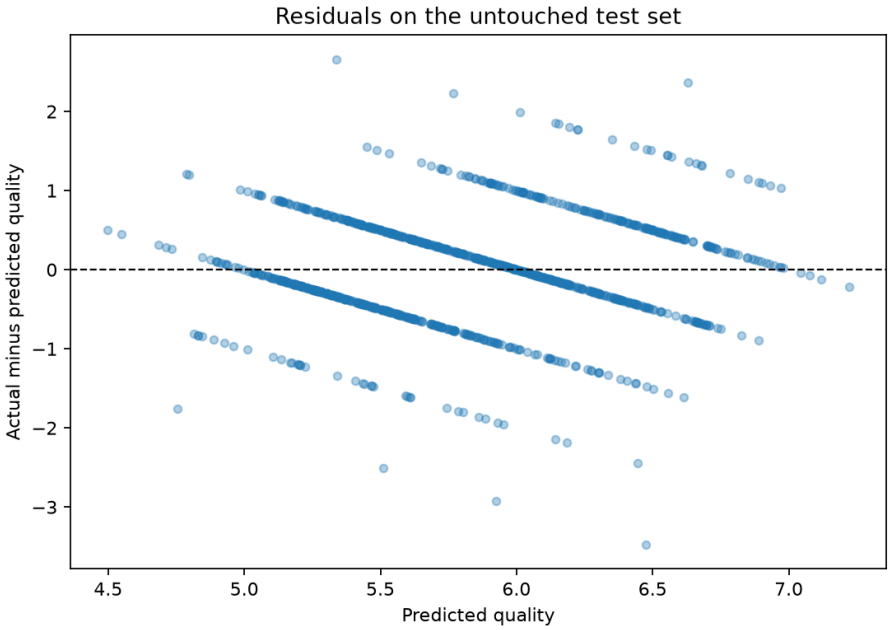


Figure 3. Residuals for the selected model on the held-out test split.

Residuals are observed scores minus predictions. Their spread shows uncertainty for individual wines; the model is not a substitute for expert tasting.

4. Interpretation and limitations

Permutation importance ranks inputs by how much test RMSE rises when an input is shuffled. Alcohol ranks highest for this fitted model. This is predictive association, not evidence that changing alcohol causes a quality change. Correlated inputs can share or obscure importance.

Feature	Mean RMSE increase	Meaning
alcohol	0.108	More reliance in this fitted model
volatile_acidity	0.039	More reliance in this fitted model
free_sulfur_dioxide	0.036	More reliance in this fitted model
sulphates	0.033	More reliance in this fitted model
total_sulfur_dioxide	0.033	More reliance in this fitted model

Limitations

- Quality is an ordinal sensory score, modelled numerically; predictions can be fractional.

- The data cover Portuguese Vinho Verde wines and may not generalize to other regions or future samples.

- The inputs omit grape variety, producer, price and tasting context; rare high scores have few examples.

- Duplicate removal may omit distinct wines that happen to share all recorded measurements.

- Correlations and feature importance do not establish causation.

Reproducibility and reference

In the project environment, install with `python -m pip install -e ".[dev]"`, then run `wine-quality train` to repeat training, selection and final evaluation. The saved model and evaluation JSON correspond to the results in this report. Review the notebook and AI usage statement before submission.

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Wine Quality [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C56S3T. Licensed CC BY 4.0.